VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

**Department of Computer Engineering**

# CAPSTONE PROJECT LOGBOOK

**(26CMCP71 : Capstone Project - II)**

**2026-27**


**GROUP MEMBERS**

**Vedant Mhatre (D17C - 32)**

**Aditya Upasani (D17C - 62)**

**Yash Mahajan (D17C - 27)**

**Sanjay Aski (D17C - 3)**

**Project Mentor**

**Dr. Sharmila Sengupta**

**Professor**

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

# Department of Computer Engineering

## INSTITUTE VISION & MISSION

**VISION:**

To build a vibrant, knowledge-driven ecosystem that promotes holistic learning, inspires innovation, practices sustainability and value-based education, and shapes socially conscious, industry-ready technocrats.

**MISSION:**

- To inculcate a culture of multidisciplinary value based education.
- To inspire students to apply their knowledge to meet the needs of industry and society.
- To foster research, entrepreneurship and lifelong learning for societal growth

## COMPUTER ENGINEERING DEPARTMENT

**VISION:**

To establish a center of excellence in computing by empowering students with advanced technological skills for fostering innovation, research, and societal impact.

**MISSION:**

- To cultivate a teaching learning environment that unites theory with computing skills, empowering students to adapt to evolving technologies.
- To strengthen core computing knowledge by fostering research, innovation, and entrepreneurship through multi-disciplinary learning.
- To prepare competent professionals with industry-ready skills, encouraging social awareness, sustainable development, and a commitment to lifelong learning.

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

# Department of Computer Engineering

## PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

| | |
|---|---|
| I | To equip Computer Engineering students with a strong foundation in mathematics, science, and core computing and engineering principles, enabling them to comprehend, analyze, and design effective technology-driven solutions. |
| II | To foster a multidisciplinary mindset in Computer Engineering students through flexible learning, enabling adaptability along with strong communication, teamwork, and leadership skills. |
| III | To empower Computer Engineering students with advanced computing and software skills, fostering innovation, entrepreneurship, and sustainable practices for lifelong learning. |
| IV | To provide Computer Engineering students with broad educational and research experiences through academic and industrial collaboration programs, preparing them for industry-ready careers. |

## PROGRAM OUTCOMES (POs)

| Program Outcome Code | Program Outcome Description |
|---|---|
| PO1 | **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems. |
| PO2 | **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4) |
| PO3 | **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) |
| PO4 | **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8). |

# Department of Computer Engineering

| | |
|---|---|
| PO5 | **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6) |
| PO6 | **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7). |
| PO7 | **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9) |
| PO8 | **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams. |
| PO9 | **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural,language, and learning differences. |
| PO10 | **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments. |
| PO11 | **Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8) |

## PROGRAM SPECIFIC OUTCOMES (PSOs)

| | |
|---|---|
| PSO1 | **Professional Skills -** The ability to develop programs for computer based systems of varying complexity and domains using standard practices. |
| PSO2 | **Successful Career -** The ability to adopt skills, languages, environment and platforms for creating innovative career paths, being successful entrepreneurs or for pursuing higher studies. |

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

## Department of Computer Engineering

## STUDENT INFORMATION

**Project Title:** A quantum-enhanced graph learning framework for edge-level traffic forecasting and QAOA-based congestion rerouting

**SDGs Mapped:** SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), SDG 13 (Climate Action)

| | Student 1 | Student 2 | Student 3 | Student 4 |
|---|---|---|---|---|
| **Roll No.** | 32 | 62 | 27 | 3 |
| **Name** | Vedant Mhatre | Aditya Upasani | Yash Mahajan | Sanjay Aski |
| **Class with Division** | CMPN - D17C | CMPN - D17C | CMPN - D17C | CMPN - D17C |
| **Contact No.** | 8779587397 | 9284946023 | 8828301189 | 8766531982 |
| **E-mail** | 2023.vedant.mhatre@ves.ac.in | 2023.aditya.upasani@ves.ac.in | 2023.yash.mahajan@ves.ac.in | 2023.sanjay.aski@ves.ac.in |
| **Address** | D2003, Sheth Zuri, Near Viviana Mall, Eastern Express Highway, Thane West 400601 | Flat 704, Tower 4, Seasons Park, Ambernath East | 8/703, Tulsidham Bhairavi CHS, Near DAV Public School, Thane West | Suresh kene Chawl R-4 C-6 Near Tulsa Park apt , bardan Galli, Khadegolvali Kalyan 421306 |

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

## Department of Computer Engineering

## INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the Project Mentor or Project Coordinator for verification and evaluation of project activities at least once in a week.
2. A logbook duly signed by the project mentor must be submitted with a project blackbook for evaluation at the end of semester to the department.

## DECLARATION

I declare that this project represents my ideas in my own words and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the Institute norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully

1. Vedant Mhatre (D17C - 32)
2. Aditya Upasani (D17C - 62)
3. Yash Mahajan (D17C - 27)
4. Sanjay Aski (D17C - 3)

(Signature of Students)

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

**Department of Computer Engineering**

# Letter of Acceptance

I undersigned, **Dr. Sharmila Sengupta**, working in the Computer Engineering department, willing to guide the Project titled "*A quantum-enhanced graph learning framework for edge-level traffic forecasting and QAOA-based congestion rerouting*" for the Capstone Project - II Semester VII respectively for the ***Academic Year 2026-27.*** The names of the students are:

1. **Vedant Mhatre**
2. **Aditya Upasani**
3. **Yash Mahajan**
4. **Sanjay Aski**

(Project Mentor) (Capstone Project Coordinator) (HOD Department of Computer Engineering)

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

# Department of Computer Engineering

## COURSE OUTCOMES FOR CAPSTONE PROJECT–II

| Course Outcome | Description | PO's Covered | PSO's Covered |
|---|---|---|---|
| CO1 | Implement and integrate all modules of the proposed system effectively. | PO1, PO2, PO3, PO5 | PSO1, PSO2 |
| CO2 | Perform testing, debugging, validation, and performance optimization of the developed solution. | PO2, PO3, PO4, PO5 | PSO1, PSO2 |
| CO3 | Analyze project outcomes using appropriate engineering tools and metrics. | PO2, PO4, PO5 | PSO1, PSO2 |
| CO4 | Demonstrate deployment readiness, usability, security, scalability, and sustainability considerations. | PO3, PO5, PO6, PO7 | PSO1, PSO2 |
| CO5 | Analyze, interpret, and present research outcomes effectively using appropriate research methodologies, analytical tools, and presentation techniques. | PO2, PO4, PO5, PO8, PO10 | PSO1, PSO2 |
| CO6 | Present and defend the complete project professionally through demonstrations, presentations, and comprehensive technical documentation. | PO8, PO9, PO10, PO11 | PSO1, PSO2 |

## CO–PO–PSO Mapping Table

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 3 | 3 | – | 3 | – | – | – | 3 | – | 2 | 3 | 3 |
| CO2 | 2 | 3 | 2 | 3 | 3 | – | – | – | 2 | – | 2 | 3 | 2 |
| CO3 | 2 | 3 | 2 | 3 | 3 | – | – | – | – | 2 | 2 | 3 | 2 |
| CO4 | 2 | 2 | 3 | 2 | 3 | 2 | 3 | 2 | – | – | 3 | 3 | 3 |
| CO5 | 2 | 3 | – | 3 | 3 | – | – | 2 | – | 3 | 2 | 2 | 2 |
| CO6 | – | – | – | – | 2 | – | – | 2 | 3 | 3 | 2 | 2 | 2 |

*Mapping scale: 3 = High correlation; 2 = Moderate correlation; 1 = Low correlation; – = No direct correlation.*

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

## Department of Computer Engineering

# SCHEDULE FOR CAPSTONE PROJECT - II

| Date | Week | Contents | Remark | Guide Sign |
|---|---|---|---|---|
| 3/08 | 1 | Theoretical formulation of QAOA | | |
| 10/08 | 2 | Analysis of research papers on quantum optimization | | |
| 17/08 | 3 | Exploring classical approaches to act as a baseline against QAOA | | |
| 24/08 | 4 | Complete mathematical formulation of QAOA and BPSO | | |
| 31/08 | 5 | Implementation of QAOA and BPSO on a 4-node topology | | |
| 7/09 | 6 | Scaling the implementation to 5-node and 6-node networks | | |
| 14/09 | 7 | Finalising the introduction and the literature survey | | |
| 21/09 | 8 | Drafting the methodology of the traffic forecasting section | | |
| 28/09 | 9 | Drafting the methodology of the optimization section | | |
| 5/10 | 10 | Conducting an ablation study and multi-seed tests to further strengthen the paper | | |
| 12/10 | 11 | Drafting and justifying the dataset, results and inference | | |
| 19/10 | 12 | Submission of the research paper to conferences and journals | | |

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

# Department of Computer Engineering

## PROGRESS / ATTENDANCE REPORT

| | |
|---|---|
| **Title of the Project:** A quantum-enhanced graph learning framework for edge-level traffic forecasting and QAOA-based congestion rerouting | |
| **Group No. 10** | **Vedant Mhatre**<br>**Aditya Upasani**<br>**Yash Mahajan**<br>**Sanjay Aski** |
| **Name of the Project Mentor: Dr. Sharmila Sengupta** | |

| Sr. No | Date | Attendance | | | | Progress/Suggestion | Mapping | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | | CO | PO | PSO |
| 1 | 3/08 | ✓ | ✓ | ✓ | ✓ | Gained theoretical knowledge of QAOA | 1 | 1,2,3,5 | 1,2 |
| 2 | 10/08 | ✓ | ✓ | ✓ | ✓ | Presented a survey of papers on quantum optimization | 1 | 1,2,3,5 | 1,2 |
| 3 | 17/08 | ✓ | ✓ | ✓ | ✓ | Finalised BPSO as a classical baseline | 1 | 1,2,3,5 | 1,2 |
| 4 | 24/08 | ✓ | ✓ | ✓ | ✓ | Completed the mathematical formulation of QAOA and BPSO | 1 | 1,2,3,5 | 1,2 |
| 5 | 31/08 | ✓ | ✓ | ✓ | ✓ | Implemented QAOA and BPSO on 4-node topology | 1,2,5 | 1,2,3,4,5, 8,10 | 1,2 |
| 6 | 7/09 | ✓ | ✓ | ✓ | ✓ | Experimented with 5 and 6-node networks | 1,2,5 | 1,2,3,4,5, 8,10 | 1,2 |
| 7 | 14/09 | ✓ | ✓ | ✓ | ✓ | Drafted the introduction and literature survey | 3,4 | 2,4,5,6,7 | 1,2 |
| 8 | 21/09 | ✓ | ✓ | ✓ | ✓ | Drafted the methodology of the traffic forecasting section | 1,3,4 | 1,2,3,4,5, 6,7 | 1,2 |
| 9 | 28/09 | ✓ | ✓ | ✓ | ✓ | Drafted the methodology of the optimization section | 1,3,4 | 1,2,3,4,5, 6,7 | 1,2 |
| 10 | 5/10 | ✓ | ✓ | ✓ | ✓ | Enhanced the paper by adding an ablation study and multi-seed tests | 4,5 | 2,3,4,5,6, 7,8,10 | 1,2 |
| 11 | 12/10 | ✓ | ✓ | ✓ | ✓ | Analysed the results and drew inference | 4,5 | 2,3,4,5,6, 7,8,10 | 1,2 |
| 12 | 19/10 | ✓ | ✓ | ✓ | ✓ | Submitted the research paper to conferences and journals | 6 | 8,9,10,11 | 1,2 |

**Sign of the Project Mentor**

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

## Department of Computer Engineering

# EXAMINER'S FEEDBACK FORM

Name of External examiner:_____

College of External examiner:_____

Name of Internal examiner:_____

Date of Examination:_____/____/_____

No. of students in project team:

Availability of separate lab for the project: Yes / No

**Student Performance Analysis** (Put Tick as per your Observation)

| Excellent (3) | Very Good (2) | Good (1) | | |
|---|---|---|---|---|
| **Sr. No.** | **Observation** | **(3)** | **(2)** | **(1)** |
| 1 | Quality of problem and Complex Engineering Problems | | | |
| 2 | Innovativeness in solutions | | | |
| 3 | Cost effectiveness and Societal impact | | | |
| 4 | Full functioning of working model as per stated requirements | | | |
| 5 | Effective use of skill sets | | | |
| 6 | Effective use of standard engineering norms | | | |
| 7 | Contribution of an individual as member or leader | | | |
| 8 | Clarity in written and oral communication | | | |
| 9 | Overall performance | | | |

Suggest new Innovative Technique/Idea/ objectives related to this project.

_____

_____

_____

**Signature of External Examiner** **Signature of Internal Examiner**